

The Well-Tempered Midwest

Reyner Banham, the ancient arts of shelter, and how to build a lean mechanical system in Chicago

I. The Argument, Standing on Dearborn Street

Before you read another word about passive design, go stand at the base of the Monadnock Building on West Jackson.

Burnham & Root finished the north half in 1891. Sixteen stories of load-bearing brick — no steel frame carrying the walls, just masonry doing the work all the way down — and at the base those walls swell to something on the order of six feet thick. It is usually taught as a structural curiosity, the last gasp of masonry before the Chicago frame made it obsolete, a dead end on the way to Mies.

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Source when available: Wikimedia Commons — search “Monadnock Building Chicago”

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Fig. 1 — *Monadnock Building, Burnham & Root, 1891. Sixteen stories of load-bearing brick; note the projecting bays pulling glazing into a narrow floor plate.*

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Source when available: Wikimedia Commons — search “Monadnock Building Chicago base” (or: “Monadnock Building entrance”)

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Fig. 2 — *The base course. The wall thickens toward the ground because the masonry is carrying the load — and incidentally forms an enormous thermal flywheel.*

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That is the wrong lesson. Look at what that building *does*. Six feet of brick is a thermal flywheel of extraordinary capacity. The floor plate is narrow, because daylight only reaches about twenty-five or thirty feet from a window and nobody in 1891 had another option. The ceilings are high, because warm air stratifies and you want it above your head. The windows operate, because there was no other way to move air. The bays project, because that gets more glass surface into a narrow plate and lets you catch air from more than one direction. Every one of those moves is a *thermal* move, made by architects who had no mechanical system worth speaking of and therefore had to make the building itself do the environmental work.

The Monadnock is Chicago's own 2226. It was standing for a hundred and twenty-two years before Baumschlager Eberle finished their famous heating-free office block in Lustenau, and it embodies most of the same principles: mass, restraint in glazing, high ceilings, occupant-operated ventilation, and a design life measured in centuries rather than decades.

We forgot how to do this. That forgetting is the subject of the single most important book on your reading list, and it is not a green building book at all.

II. Banham's Real Thesis

Reyner Banham published *The Architecture of the Well-Tempered Environment* in 1969. If you know one thing about it, it is probably the accusation: that the history of architecture has been written as a history of structure and form — of walls and columns and façades — while the machinery that actually makes buildings habitable was excluded from the story entirely. Architectural historians wrote about the Crystal Palace's iron and glass and said nothing about how it handled solar gain. They wrote about the modern office tower and ignored the fact that it could not exist without a boiler, a chiller, and a fan.

Banham's fix was to write the other history: environmental management as a *primary* architectural act rather than a technical afterthought. And in doing so he gave us the taxonomy that I think should organize your entire inquiry.

He proposed that human beings have only ever had three ways to make a hostile environment habitable.

The conservative mode. This is the cave, the pit house, the massive wall, the courtyard house buried in its own thermal mass. You build something heavy and enclosing, and it *stores* — it takes in heat during the day and gives it back at night, it damps the swings, it holds the average. The structure and the environmental system are the same object. There is no distinction between the thing that holds the roof up and the thing that keeps you comfortable. Thick masonry, rammed earth, the Anasazi cliff dwelling, the Roman insula, the Monadnock, 2226 Lustenau.

The selective mode. This is the tent, the screen, the shutter, the veranda, the brise-soleil, the *mashrabiya*. Instead of storing, you *filter* — you admit what you want (light, breeze, winter sun) and exclude what you don't (glare, rain, summer sun, insects). The membrane is thin and smart rather than thick and dumb. Japanese domestic architecture is the canonical case; so is the Southeast Asian house on stilts with deep eaves; so, in the modern era, is every well-designed shading system and operable façade.

The regenerative mode. This is the campfire. You bring *power* to the problem. You burn something, or you run a compressor, and you generate the environment you want directly, without asking the building fabric to help. This is the mode that industrial civilization fell in love with, and it is the mode that produced the sealed glass box, because once you have enough power you no longer need the building to do anything environmental at all.

Every building in history is some mixture of the three. What changed in the twentieth century is that the price of the regenerative mode collapsed, and so it ate the other two.

That is the frame. Hold onto it, because it turns "how do we not make HVAC so complicated" into a precise engineering question: **how much of the environmental work can I push back into the conservative and selective modes, so that the regenerative mode has less to do?**

III. What Banham Actually Wanted (and the Misreading Everyone Commits)

Here is where you should be careful, because Banham gets conscripted constantly into a sustainability argument he did not quite make, and you will sound sharper if you know the difference.

Banham was not a nostalgist. He was not arguing that we should go back to thick walls and open windows. He was, if anything, thrilled by the American power-operated environment — by the audacity of ducting and refrigerating and illuminating your way out of climate. His sympathies ran toward Buckminster Fuller and the technological sublime, not toward vernacular masonry. Four years before the book, he published "A Home Is Not a House," illustrated by François Dallegret, which proposed the "Environment Bubble": a transparent inflatable dome with a machine at its center providing heat, light, sound, and power. The joke — and it was only half a joke — was that if the machinery is what actually produces the environment, then the machinery *is* the architecture, and the house is just decorative cladding we keep around for sentimental reasons.

So Banham's complaint was not "too much machinery." It was **dishonesty about the machinery**. His charge was that architects had abdicated: they handed the environmental problem to engineers, refused to look at what came back, hid the ducts above the ceiling, and then went on writing about form as though form were the whole story. The result was buildings whose architecture was environmentally inert — which made heroic machinery *necessary* — and whose machinery was then treated as beneath discussion.

Read that way, Banham is not an argument for eliminating HVAC. He is an argument for **designing the building and the mechanical system as one thing**. Which is exactly what Baumschlager Eberle did, and exactly what a typical American K-12 school does not do.

There's a second lesson buried in his history that matters even more for your purposes. The moment the regenerative mode became cheap, the *constraints* that had shaped good buildings disappeared — and with them, the buildings. As long as you needed daylight, floor plates had to be narrow. As long as you needed cross-ventilation, you needed operable windows on two sides. As long as heat was expensive, you needed compact form and modest glazing. Those constraints produced the Rookery's light court, the Monadnock's slender bar, the Marquette's narrow wings, and every light well in every prewar Chicago building. They were not aesthetic preferences. They were the physics showing through.

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Source when available: Wikimedia Commons — search "Rookery Building Chicago light court"

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Fig. 3 — The Rookery's light court. Daylight and cross-ventilation were the binding constraints; the light well is the physics showing through.

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Fluorescent lighting and cheap refrigeration dissolved them all. You could suddenly build a hundred-and-fifty-foot-deep floor plate with a sealed skin, light it artificially, and cool it mechanically, and nothing in the market punished you for it. Chicago's own architectural history is the cleanest possible illustration of

Banham's thesis: the city invented the tall office building under environmental constraint, then invented the sealed curtain wall once the constraint was lifted, and the second invention conquered the world.

The passive project, properly understood, is not a return to 1891. It is the deliberate re-imposition of those constraints — chosen this time rather than suffered — because we have learned that the constraints were producing something valuable.

IV. The Wright Detour: Chicago Was There First, Both Times

There's an intermediate figure worth knowing, and Banham spends real time on him: Frank Lloyd Wright's Larkin Administration Building in Buffalo, completed around 1906.

Banham's interest in the Larkin is that it was an early and unusually complete attempt to *integrate* mechanical services into architectural form rather than bury them. The building was sealed against a filthy industrial railyard environment, and its air was drawn in, filtered, tempered, and distributed — with the ductwork and services housed in the great corner towers that give the building its unmistakable massing. The environmental system generated the architecture. The towers are not decorative; they are the plant. (The frequent claim that the Larkin was "the first air-conditioned office building" is contested and probably overstated — what it certainly had was engineered, filtered, tempered mechanical ventilation, which was radical enough for 1906.)

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Source when available: Wikimedia Commons — search "Larkin Administration Building"

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Fig. 4 — *Larkin Administration Building, Frank Lloyd Wright, Buffalo, c.1906. The corner towers are the air-handling plant.*

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Source when available: Wikimedia Commons — search "Larkin Building interior"

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Fig. 5 — *The top-lit central court. Sealed against the railyard, mechanically ventilated, filtered and tempered.*

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Then Wright did the opposite thing, and did it just as well. Unity Temple, Oak Park, 1908: a heavy concrete box with a top-lit sanctuary, ventilation moving through the hollow structure itself. The Johnson Wax Administration Building, 1939, sealed and glass-tubed and lit from above. Wright oscillated between conservative-mode mass and regenerative-mode machinery his entire career, sometimes in the same building.

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Source when available: Wikimedia Commons — search "Unity Temple Oak Park interior"

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Fig. 6 — *Unity Temple. Wright working the conservative mode: heavy concrete, top light, ventilation moving through the structure itself.*

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The point of the detour is this: nobody has to invent the integration of architecture and environmental system for the Midwest. It was already done here, twice, by the two most consequential architectural cultures the region produced. What was lost was not the knowledge. It was the *reason to bother*.

V. Why Lustenau and Harare Do Not Ship to Chicago

Now the hard part, and the part that will keep you honest in front of an engineer.

The two great "no mechanical systems" buildings — 2226 in Lustenau, Austria, and Mick Pearce's Eastgate Centre in Harare — are real, they work, and their strategies are largely **untransferable to Chicago**. Not because the physics is different, but because the climate does something specific that defeats them.

FIGURE 7 — image not embedded

Source when available: Wikimedia Commons — search "2226 Lustenau Baumschlager Eberle"

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Fig. 7 — 2226, Baumschlager Eberle, Lustenau, 2013. Deep-set punched windows in an ~80cm plastered brick wall. No heating, no cooling, no mechanical ventilation.

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Both buildings run on the same trick: **thermal mass charged by night flushing**. You let the building soak up heat all day, and at night you push cool outdoor air through it to dump that heat, so you start the next day cold. The trick has two prerequisites, and they are non-negotiable:

1. **A large diurnal swing.** The night has to get meaningfully colder than the day. The usual design rule of thumb is that you need something like a 20°F daily swing, with nighttime temperatures running roughly 10–12°F below your target indoor daytime temperature.
2. **Dry night air.** You are pushing outdoor air across interior surfaces. If that air is loaded with moisture, you are not cooling the building — you are wetting it.

Harare sits at nearly 5,000 feet on a high plateau with enormous diurnal swings and a genuinely dry season. Lustenau is in the alpine foreland with July nights that fall into the mid-to-upper 50s and low dewpoints. Both of those places hand you the prerequisites for free.

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Source when available: Wikimedia Commons — search "Eastgate Centre Harare"

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Fig. 9 — Eastgate Centre, Mick Pearce with Arup, Harare, 1996. The rooftop stacks are the buoyancy engine driving night flush through the concrete mass.

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Chicago does not. Roughly speaking, a Chicago July gives you daytime highs in the mid-80s and overnight lows around 68–70°F. That's a swing of maybe 15–18°F — already marginal. But the killer is the moisture. Mean July dewpoints here run in the low-to-mid 60s, with regular multi-day stretches at 70–75°F, and the ASHRAE summer design dewpoint is up around 74–75°F. Overnight, as the dry-bulb temperature falls toward the dewpoint, relative humidity climbs toward saturation. **The cool night air you want to flush through your building is, in a Chicago July, nearly saturated air.** Push it across a slab you have chilled below its dewpoint and you get condensation, and then you get mold, and then you get a lawsuit.

This is the single most important thing to understand about applying European and tropical-highland passive strategies in the American Midwest, and it explains a persistent transatlantic misunderstanding. European building scientists treat night flushing as a general-purpose cooling solution because European summer dewpoints are substantially lower than eastern American ones. It works there. Applied here without modification, it is a moisture-damage mechanism.

The deeper structural fact is this: **as you superinsulate and tighten a building, the sensible cooling load collapses but the latent load does not.** Sensible load comes from conduction and solar gain through the envelope, and you can attack both with insulation and shading. Latent load comes from outdoor air ventilation and from people, and ventilation rates are set by code and occupancy, not by your envelope. So the better your envelope gets, the *lower* your sensible heat ratio goes — and conventional direct-expansion cooling equipment cannot handle a very low sensible heat ratio. It removes too much temperature and not enough moisture. Then, because your loads are so small, the equipment short-cycles: it satisfies the thermostat in six minutes, shuts off before the coil has stayed wet long enough to strip meaningful moisture out of the air, and leaves you with a building that is 72°F and 65% relative humidity. Cold and clammy. This is the classic failure mode of low-load buildings in humid climates, and it is thoroughly documented.

So: in Chicago, humidity is the boss. Any strategy that ignores latent load is not a strategy.

But note what Chicago *does* give you

Having said all that, do not conclude that Chicago is hostile to this project. It gives you two enormous gifts.

Winter. Chicago is a heating-dominated climate — on the order of 6,000-plus heating degree days against maybe 900–1,000 cooling degree days. The conservative mode — mass, compactness, superinsulation, airtightness, solar gain — is *most* effective exactly where the loads are biggest. Everything Passive House does well, Chicago rewards. There is no latent problem in January; there is a desiccation problem, which is the opposite and easier.

The shoulders. Chicago has genuinely excellent spring and fall. There are thousands of hours a year when outdoor air is between roughly 50°F and 68°F with dewpoints below 55°F — which is to say, thousands of hours when outdoor air is *free cooling*, requiring nothing but an open damper or an open window. Any building that cannot exploit those hours is throwing away the best gift this climate offers.

And if you are working with school districts, notice the schedule. The Chicago-area school year runs roughly from late August to early June. **The two worst latent-load months, July and August, are the two months the buildings are least occupied.** That is not a small detail — it materially changes what "adequate cooling" means for a K-12 building relative to an office or an apartment tower. You still need real capacity for late August, September, May, and June. But the design peak you must chase is meaningfully narrower than the calendar suggests, and the shoulder-season free-cooling hours land squarely inside the occupied season.

VI. Sorting the Ancient Techniques: What Survives Translation

Here is the ancient toolkit, sorted honestly by what actually works in Zone 5A.

Works, essentially unmodified

The hypocaust → the radiant slab. The Romans ran hot flue gas through a raised floor and heated the room from its surfaces rather than its air. That is precisely what a hydronic radiant slab does, with water instead of smoke. The lineage is direct and unbroken, and it is the single most useful ancient technique available to you. It works because radiant heat transfer is what the human body actually responds to — Heschong's point, and the physiological reason a cold room with a warm floor feels better than a warm room with a cold floor. It also works because it is *low-exergy*: you heat with 90–110°F water instead of 180°F water, which lets condensing boilers condense and lets heat pumps run at high coefficient of performance. Chicago's heating-dominated climate makes this a straightforward win.

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Fig. 10 — Hypocaust, Roman. Raised floor on pilae, hot gas circulating beneath, the room heated from its surfaces.

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Fig. 14 — Hydronic tubing before pour. The hypocaust with a pump.

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Mass, orientation, compactness, and shading. All still good. External shading in particular is criminally underused in American commercial work. The summer sun in Chicago is high — a horizontal overhang on a south face is extremely effective and cheap; east and west faces need vertical fins or, better, less glass. Deep window reveals do real work for free. Reducing window-to-wall ratio from the reflexive 60–70% to something like 30–40%, with better glass and real shading, is the single highest-leverage architectural decision in the whole project. It is also what makes the Monadnock and 2226 look the way they look.

The cellar → ground coupling. Below about twenty feet, Chicago soil sits around 50–55°F year-round. That is a magnificent heat sink in summer and a magnificent heat source in winter. Modern translation: ground-source heat pumps and geo-exchange fields. This is the ancient technique most worth the capital.

The hearth → thermal variety. Heschong's *Thermal Delight in Architecture* is the theory: the modern insistence on uniform temperature everywhere is a recent, impoverished idea, and people are demonstrably comfortable across a much wider band when they have local control, radiant sources, air movement, and variation. Practically, this means ceiling fans, operable windows, sunny alcoves, and the willingness to let a building be 68°F in one place and 76°F in another. Widening the acceptable comfort

band is the cheapest cooling capacity you will ever buy — the adaptive comfort model in ASHRAE 55 exists precisely to let you claim it.

Works, but only with modification

Night flushing. Not dead — just seasonal. In Chicago it is a *shoulder-season* strategy: genuinely powerful in May, early June, September, and October, and unreliable-to-harmful in July and August. Design it with a dewpoint interlock, not just a dry-bulb interlock. The control logic is: flush when outdoor dry-bulb is below the indoor setpoint *and* outdoor dewpoint is below roughly 55–58°F. Simple, cheap, and it will run for hundreds of hours a year.

The windcatcher and the stack. Badgirs, solar chimneys, and stack ventilation all work on buoyancy and wind pressure, and the physics is climate-independent. What is *not* climate-independent is the Persian version's habit of coupling the windcatcher to a qanat or a fountain, so the incoming air is evaporatively cooled. **Evaporative cooling is dead on arrival in Chicago** — you cannot cool by adding moisture to air that is already wet. So take the stack, leave the water. Solar chimneys and atria driving buoyant exhaust remain useful here, particularly in shoulder seasons and particularly for schools, where stack ventilation through a central atrium is architecturally natural.

▽ **FIGURE 11 — image not embedded**

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Fig. 11 — Badgirs at Yazd. Take the tower; leave the qanat. Buoyancy travels to Chicago, evaporative cooling does not.

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Exposed thermal mass. Useful, but it fights the rest of the building. To get mass working you need exposed slab soffits, which means no dropped ceiling, which means your services and your acoustics have nowhere to hide. In a classroom, acoustics is not optional — reverberation control is a genuine learning-outcome issue. This is a real, unresolved design tension, not something to wave away. Thermally activated building systems (TABS — pipes cast into the slab) are the sophisticated resolution, and they are common in Europe and rare here.

Dead on arrival in Chicago

Earth tubes. Running ventilation air through buried pipes to precool it is an ancient and beautiful idea. In a humid continental climate it is a mold farm. You are pulling 75°F dewpoint air through a 55°F pipe; it condenses inside the tube, the tube stays wet, and now you are ducting biological growth into occupied space. The correct modern translation is: **couple to the ground with water, not with air**. Same physics, no condensation in an inaccessible pipe.

Pure evaporative cooling in any form — direct or indirect-direct. Wrong climate.

The unmodified 2226 model — no heating, no cooling, no mechanical ventilation, comfort held by mass and occupant-operated vents alone. It is a beautiful building and a genuine achievement, but it depends on a mild alpine climate, low occupant density, and an occupant culture willing to accept a 22–26°C band. It is also, worth noting, not a Passive House at all: it has no heat recovery ventilation. In Chicago, ASHRAE 62.1 minimum ventilation rates alone will hand you a latent load that no amount of brick can absorb.

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Fig. 8 — The sensor-controlled ventilation flap. The entire “mechanical system,” essentially.

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VII. The Lean Machine: What to Actually Build in Chicago

So what does "not making HVAC so complicated" look like when you write it into a basis of design for a Midwestern school, municipal building, or apartment block?

The answer has a name, and it is not "passive." It is **envelope-first, then decoupled**. Here is the architecture, in the order the decisions must be made.

1. Spend the money on the envelope, once

Airtightness first — it is the cheapest and most permanent performance you will ever buy, and it is the prerequisite for everything downstream. Passive House asks 0.6 ACH50; even without certification, a commercial target in the range of 0.15–0.25 cfm/sf at 75 Pa is achievable and transformative. Add continuous exterior insulation with thermal bridges actually detailed out, high-performance glazing with solar heat gain coefficient tuned by orientation, external shading, and a disciplined window-to-wall ratio.

The reason this comes first is not virtue. It is that **the envelope determines the size and complexity of everything else**. A tight, well-insulated, well-shaded building has small loads, and small loads are what make simple mechanical systems possible. You cannot buy simplicity in the mechanical room; you can only earn it in the wall section.

2. Decouple ventilation from thermal conditioning

This is the central move, and it is the direct descendant of Banham's argument. In a conventional variable-air-volume system, one medium — air — is asked to do three unrelated jobs at once: deliver oxygen and dilute contaminants, deliver or remove sensible heat, and remove moisture. Because air is a lousy heat-transport medium, you must move enormous quantities of it to do the thermal job, which means big ducts, big fans, big shafts, and a box with a damper and a reheat coil in every zone. The complexity you object to is almost entirely a consequence of using air as a heating and cooling fluid.

So stop. **Split the jobs by medium:**

- **A dedicated outdoor air system (DOAS)** handles ventilation and *all* of the latent load. It delivers a small, constant, code-driven volume of outdoor air, conditioned to a low dewpoint.

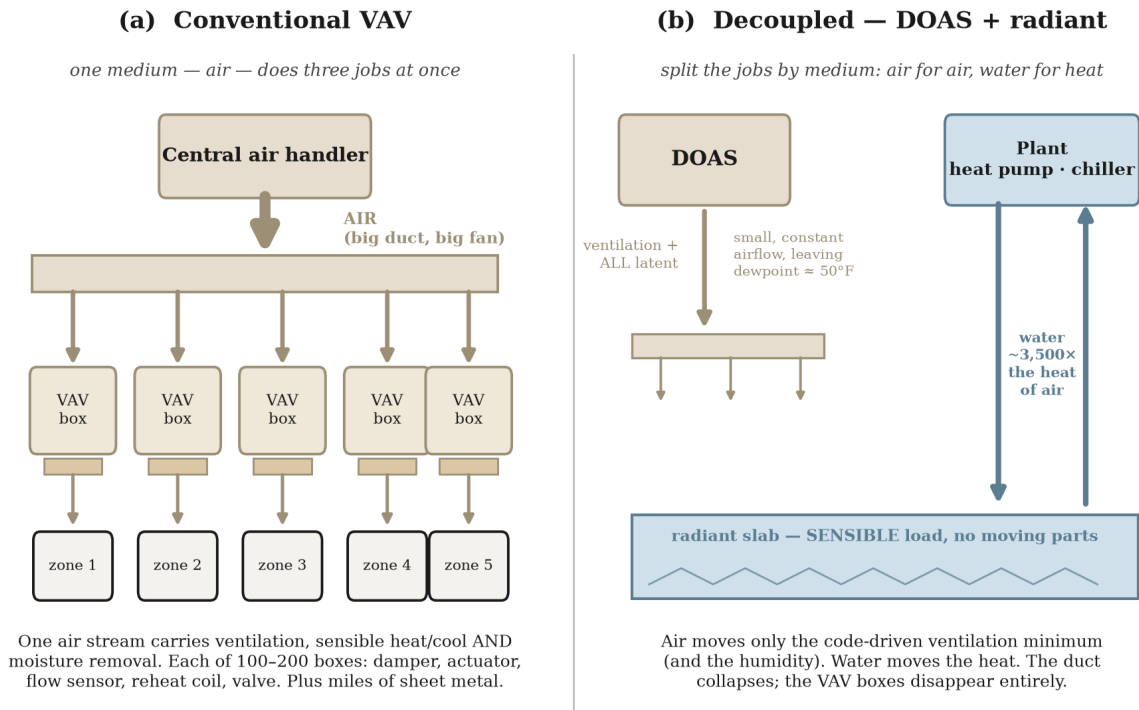


Fig. 12 — Decoupled architecture: DOAS carries ventilation and all latent load; hydronic radiant carries sensible load.

Diagram prepared for this document

- **Hydronic radiant surfaces** — slab, panel, or TABS — handle the sensible load. Water carries roughly 3,500 times the heat per unit volume that air does, so the distribution shrinks to pipes.

The duct system collapses to a fraction of its former size. The VAV boxes disappear entirely. This is the hypocaust and the Roman bath, rebuilt with a pump.

3. Make the DOAS do the humidity work, properly

In Chicago the DOAS is where the fight is won or lost. Requirements:

Deep dehumidification. Cool the outdoor air stream down to a leaving dewpoint around 50–52°F. That single stream then absorbs the entire building latent load, and every other space-conditioning device is freed to do pure sensible work.

Reheat without new energy. Air leaving a 50°F coil is too cold to deliver directly. Do not burn fuel or run resistance heat to warm it back up — use a wraparound heat pipe or a passive dehumidification wheel, which takes the heat you just removed and puts it back on the leaving side for free. This detail is the difference between a DOAS that is efficient and one that is an embarrassment.

Energy recovery, and choose the type deliberately. Chicago has both a cold, dry winter and a hot, humid summer, which generally argues for an enthalpy (total energy) device — an ERV — over a sensible-only HRV. In summer it pre-dries incoming air using the exhaust stream; in winter it retains indoor moisture and keeps you off the desiccation floor. Two cautions: understand your frost-control strategy below about 5°F, and understand the cross-contamination characteristics of a wheel if the

application is sensitive. And be clear on the limit — **an ERV is not a dehumidifier**. When outdoor air is at 90% relative humidity, an ERV that removes a fifth of the incoming moisture still raises your indoor humidity. It reduces the dehumidification load; it does not eliminate it.

Keep interior surface relative humidity below 70%, and never let it sit above 60% for extended periods. That is the mold line, and it is the reason the DOAS is not optional.

4. Interlock the radiant system to dewpoint — this is the one that bites

Radiant cooling in a humid climate has exactly one catastrophic failure mode: you chill a surface below the space dewpoint and it sweats. The control logic must be absolute and hard-wired:

- Space dewpoint sensors, not just humidity sensors.
- Chilled water supply temperature reset upward as space dewpoint rises.
- Slab surface temperature held a safe margin — typically 2–3°F — above measured space dewpoint at all times.
- **The DOAS leads.** Radiant cooling is not permitted to operate until the DOAS has established and is maintaining the low space dewpoint. On DOAS failure, radiant cooling locks out.

Get this right and you can run a slab at 62–65°F all summer in Chicago and never see a drop of condensate. Get it wrong once and you will spend the next decade explaining it.

5. Mixed mode, on the shoulders, with the changeover honest

Add natural ventilation — but as **changeover mixed mode**, not concurrent. Concurrent mixed mode (windows open while the mechanical system runs) is how buildings end up conditioning the outdoors. Changeover means the building is either in mechanical mode or in natural mode, and the controls enforce it.

The implementation that actually works: automated window or louver actuators on the building management system for the primary openings, plus manual operable windows for occupant agency, plus a simple visible indicator in each space — a green light meaning "open your windows now," red meaning "we're on mechanical, please close them." The indicator light is not a gimmick. Occupant behavior is the largest uncontrolled variable in every mixed-mode building ever measured, and telling people plainly what the building wants is the cheapest control system in the catalog.

Use this for the Chicago shoulders. Between the free-cooling hours and the night-flush hours, you are looking at a large fraction of the occupied school year running on an open damper.

6. Simplify the count

Now step back and count devices, because this is where the argument lands for your day job.

A conventional VAV system in a 100,000 square foot school: one or two large air handlers, plus something on the order of 100 to 200 terminal boxes, each with a damper, an actuator, a controller, a flow sensor, and very often a reheat coil with its own valve. Plus miles of sheet metal. Plus the outdoor air economizer dampers on the AHUs. Every one of those is a device with a service life, a failure mode, a spare parts inventory, a maintenance task, and a line in your capital plan.

A single VAV terminal box

one of 100–200 in a typical 100,000 sf school

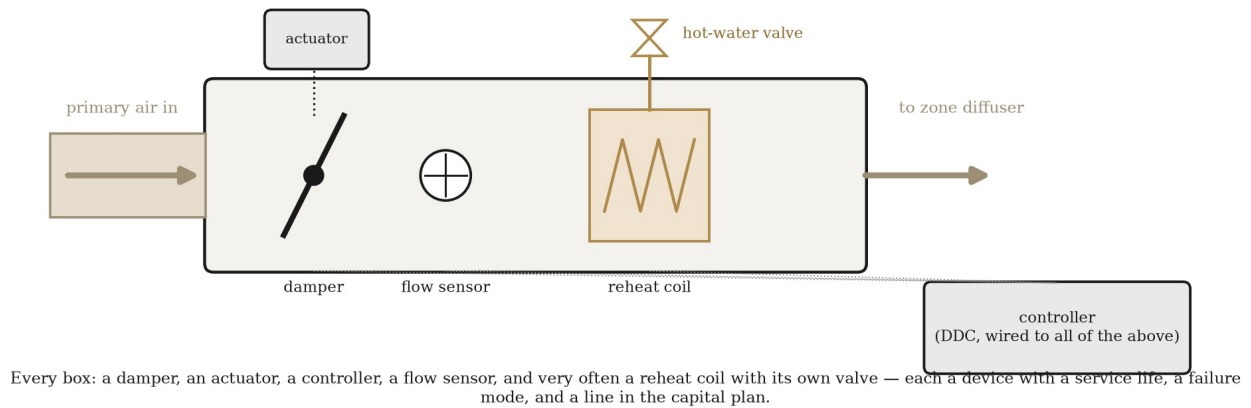


Fig. 13 — A VAV terminal box: damper, actuator, controller, flow sensor, reheat coil, valve. Now multiply by 150.

Diagram prepared for this document

The decoupled system: one or two DOAS units. A hydronic loop with pumps and valves. A slab. That is the entire thermal distribution system, and the slab has no moving parts at all.

There is a real first-cost premium, and you should quote it honestly. The best-documented comparison — a University of California Berkeley Center for the Built Environment study with Taylor Engineering — put radiant plus DOAS at roughly \$38.9 per square foot against \$29.9 for VAV, a premium around \$9.0/sf in high-labor Bay Area conditions and closer to \$6.8/sf at national average labor rates. The premium is driven almost entirely by hydronic piping labor; notably, the sheet metal cost went *down* substantially, which is exactly what you'd predict. Lifecycle cost over 20 to 30 years generally favors the decoupled system, driven by fan energy, plant downsizing, equipment count, and service calls.

VIII. Shearing Layers as a Mechanical Strategy

Now here is the synthesis I think is genuinely yours to make, and it connects Banham to Stewart Brand to your own FCA practice.

Brand's *How Buildings Learn* — borrowing from Frank Duffy — organizes a building into layers that change at radically different rates: Site (eternal), Structure (100+ years), Skin (20–50), Services (10–20), Space plan (3–10), Stuff (daily). Duffy's original formulation is the sharper one: there is no such thing as *a building*; there are several layers of longevity temporarily occupying the same coordinates. The design imperative that follows is that the layers must be able to slip past one another — you must be able to replace the fast layers without demolishing the slow ones.

Apply that to environmental design and something clarifies immediately.

Banham's conservative and selective modes live in the Structure and Skin layers. The regenerative mode lives in the Services layer. Mass, orientation, compactness, floor plate depth, ceiling height, shading geometry, and window-to-wall ratio are all hundred-year decisions embedded in the slowest layers. Boilers, chillers, fans, VAV boxes, and controls are fifteen-to-twenty-year decisions in the fastest structural layer.

Which means the strategy states itself: **push as much environmental function as possible into the slow layers, and keep the fast layer small, simple, and accessible.**

This is exactly what Baumschlager Eberle meant with their slogan about good architecture lasting a hundred years while building services last maybe twenty. It is what Alex Gordon meant in 1972 with "long life, loose fit, low energy" — three properties that turn out to be one property viewed from three sides. And it is, precisely, a capital planning strategy.

Consider what it does to a facility condition assessment. In a conventional building, the mechanical system is an enormous, distributed, fast-decaying asset class. Hundreds of components on fifteen-to-twenty-five year cycles, all of them buried above ceilings, all of them contributing to your facility condition index, all of them coming due at once because they were all installed at once. The twenty-year mechanical replacement event is the single largest recurring shock in a school district's capital plan, and it recurs forever.

In the lean building, the thermal function has largely migrated into the structure. **A radiant slab's service life is the structure's service life.** It is concrete and inert plastic pipe; it does not have a service life in the FCA sense. The mass has no service life. The orientation has no service life. The shading has a long one. What remains on a twenty-year cycle is a small, centralized, accessible plant.

You have not eliminated the twenty-year event. You have made it dramatically smaller, and you have moved the bulk of the environmental capability into the layer that was going to last a century anyway. That is a different capital plan — flatter, more predictable, less prone to the deferred-maintenance death spiral where the mechanical replacement is so large it gets deferred, and the deferral degrades everything else.

I have not seen this articulated well in the practitioner literature, and I think there is a genuine contribution sitting here: **organizing FCA condition indices and capital reserve schedules by shearing layer**, and then evaluating design alternatives by which layer they load. A mechanical system should be judged not only on first cost and energy but on *which layer it puts the function in*.

IX. Why Commissioning Becomes Load-Bearing

One more thing, and it is the thing that will make or break every project in this essay.

Lean systems have less margin for error than fat ones. That is not a complaint; it is arithmetic. An oversized VAV system with 40% excess capacity is a machine for papering over design mistakes, construction defects, and operational incompetence by brute force. It burns money doing it, but it holds setpoint. A decoupled DOAS-and-radiant system sized honestly against a superinsulated envelope has no such reserve. When the controls sequence is wrong, you find out.

The evidence on this is consistent and slightly humbling:

- **The House at Cornell Tech** — a 26-story Passive House tower with superb measured airtightness, 0.13 ACH50 — used roughly *twice* its modeled primary energy in its first year. The causes were not envelope failures. They were occupant behavior, plug and amenity loads, resident turnover, and the fact that the modeling tool's assumptions about how people would operate windows did not survive contact with a noisy bridge outside.
- **Council House 2 in Melbourne**, Mick Pearce's celebrated mixed-mode building, was found in a later energy assessment to be under-performing its design intent and required a full retro-commissioning effort — explicitly framed as an attempt to understand the gap between design and operation.
- Analyses of Passive House projects that missed their targets trace the misses repeatedly to skipped construction-phase quality assurance and mechanical commissioning, not to envelope theory.

And yet the same body of evidence shows that when these buildings *are* properly delivered, the savings are enormous and real: certified Passive House multifamily in Boston measuring around 21 kBtu/sf/yr against a non-certified peer group at 56; New York studies showing 32–58% measured reductions; the Ken Soble Tower retrofit in Hamilton cutting heating energy demand by 91%.

The gap between those two sets of outcomes is almost entirely commissioning, controls tuning, and measurement and verification.

Which is to say: **the discipline you already practice is the load-bearing element of the entire strategy.** Not an add-on service. The thing that determines whether any of it works. A passive or lean building without rigorous NCCx, functional performance testing of the changeover and dewpoint-interlock sequences, and a full first-year measured-versus-modeled reconciliation is not a passive building. It is a gamble with a nice wall section.

If you want the single most defensible professional position at the intersection of everything in this essay, it is that one.

X. Where to Read Next

Start here, in this order:

1. **Reyner Banham, *The Architecture of the Well-Tempered Environment* (1969; 2nd ed. 1984).** The frame for everything. Read it for the three modes and for the argument that environmental management is a primary architectural act. Read it skeptically as a green text — it isn't one.
2. **Lisa Heschong, *Thermal Delight in Architecture* (MIT Press, 1979).** Under eighty pages. The bridge from building physics to Alexander — why thermal variety and radiant sources feel better than uniform conditioned air, and why the pursuit of thermal uniformity was a mistake.
3. **Stewart Brand, *How Buildings Learn* (1994).** For shearing layers, and for the whole argument about buildings as processes rather than objects.
4. **Victor Olgyay, *Design with Climate* (1963).** The founding text of climate-responsive design. Dated in its data, permanent in its method.
5. **Baruch Givoni, *Man, Climate and Architecture and Passive and Low-Energy Cooling of Buildings*.** The quantitative backbone. This is where you go to check whether an idea actually holds up thermally before you propose it.

For the mechanical design specifically:

- **CIBSE AM13, *Mixed Mode Ventilation***, plus CIBSE Guide A (environmental design) and Guide B2 (ventilation and ductwork). The British guidance on mixed mode is far more mature than the American equivalent.
- **ASHRAE Standard 62.1** for the ventilation floor you cannot design under, and **ASHRAE Standard 55** for the adaptive comfort model that lets you widen the comfort band legitimately.
- **Building Science Corporation** (Lstiburek and Straube) on dehumidification in humid climates and on ERV/HRV selection. Their Building America reports on supplemental dehumidification are the definitive treatment of the low-load humidity failure mode.
- The **Berkeley CBE / Taylor Engineering radiant-versus-VAV cost study** for the numbers you'll need in a value engineering meeting.

For the Chicago-specific work:

Pull the actual ASHRAE Fundamentals design conditions for O'Hare and Midway rather than trusting any rule of thumb — including mine. Then build a simple bin analysis of your own: hours per year by dry-bulb *and* dewpoint. That single spreadsheet will tell you exactly how many hours of free cooling and viable night flush this climate offers, which is the number every strategy in this essay ultimately depends on. I suspect you will be surprised how large it is, and equally surprised how sharply it collapses in July.

Coda

Banham's accusation was that architects stopped taking responsibility for the environmental performance of what they drew, and handed it to a discipline they then declined to read.

The interesting thing about your position is that you are standing on the other side of that divide, in the discipline that inherited the problem, looking back. You do commissioning and condition assessment and capital planning — which is to say, you deal professionally with the consequences of a century of buildings designed as though the machinery would take care of it.

The synthesis available from where you stand is not available to most architects and not available to most engineers. It is this: **the environmental system, the maintenance burden, the capital plan, and the building's chance of being loved and kept for two hundred years are all the same problem.** A building that does its own thermal work needs less machinery; less machinery means a smaller fast-decaying asset class; a smaller asset class means a flatter capital plan and less deferred maintenance; less deferred maintenance means the building stays good; and a building that stays good, and is beautiful, is a building nobody demolishes.

That is Gordon's three L's, Banham's three modes, Brand's six layers, and Alexander's wholeness, all describing one thing from different angles. The Monadnock has been standing on Dearborn for a hundred and thirty-five years with six feet of brick doing most of the work. It is not a curiosity. It is a brief.

* * *

Prepared as a reading essay. Figures for Chicago climate are approximate and intended for orientation; use ASHRAE Fundamentals design data for any actual work. Case-study performance claims are flagged in the underlying research as measured, modeled, or claimed — the distinction matters, and the modeled numbers should not be quoted as achieved.